

About:

Specialize in the genetic improvement of forest tree species, focusing on productivity enhancement and the assessment of population genetic parameters to support tree improvement and germplasm conservation. My work also extends to Eco-rehabilitation, specifically for coal mine landscapes through forestry-based interventions.

Curriculum vitae**Sh. Manoj Gowda M****Division of Genetics and Tree Improvement****ICFRE- Institute of Forest Productivity****Lalgutwa, Kathalmpore, Ranchi 835303****manojgowdam95@gmail.com; manoj.ga3326@gov.in****Phone No: 8722495523**

Media	Citations	Hi Index	I Index
Google Scholar https://scholar.google.com/citations?user=h3k6Gs4AAAAJ&hl=en	129	4	3
Research gate https://www.researchgate.net/profile/Manoj-Gowda-M/research	138	5h-ind ex	

Experience: 2 Years + (Since 25th September, 2023)**Domain of Research/Consultancy**

- ❖ Tree Improvement
- ❖ Molecular Breeding
- ❖ Genetic Diversity estimation and Population genetics
- ❖ Eco-Rehabilitation
- ❖ Data analysis: Experimental designs & data analysis

Educational Qualification:

Degree	Year	Grade
Doctor of Philosophy Division of Genetics ICAR-Indian Agricultural Research Institute Pusa campus, New Delhi 110012	2019-present	---
Master of Science Division of Genetics ICAR-Indian Agricultural Research Institute Pusa campus, New Delhi 110012	2017-2019	9.10
Bachelor of Science (Agriculture) University of Agricultural Sciences Dharwad, Karnataka 580005	2013-2017	8.96

RESEARCH EXPERIENCE:

- ❖ **Genetic Diversity estimation of *Gmelina*, *Dalbergia* and Poplar**
- ❖ **Eco-rehabilitation of Coal mine areas through forestry intervention**
- ❖ **PhD Research:** GWAS and genomic predictions for enhancing genetic gain in rice
- ❖ **MSc Research:** Morphological and biochemical characterization of MAS-derived maize genotypes possessing ZmVTE4, crtRB1, lcyE and opaque2 genes

TRAINING/WEBINARS:

- ❖ Establishment and Data Analysis of Breeding Trials for Clone Development and Release
- ❖ Advance training in remote sensing and GIS (Hyper spectral remote sensing vegetation analysis), GIS in water resources application in forestry etc
- ❖ Genome-wide association studies and its application in agriculture (Training)
- ❖ Data science and machine learning for bioinformatics with R (Training)
- ❖ How accelerate genetic gains in crop improvement programs (Webinar)
- ❖ Application of omics in climate smart agriculture (Webinar)

❖ National seed system and policy perspectives (Webinar)

PUBLICATIONS

1. **Gowda, M.M.**, Muthusamy, V., Chhabra, R., Duo, H., Pal, S., Gain, N., Katral, A., Kasana, R.K., Zunjare, R.U. and Hossain, F., 2025. Development and Validation of Multiplex-PCR Assay for β -Carotene hydroxylase and γ -Tocopherol methyl transferase Genes Governing Enhanced Multivitamins in Maize for Its Application in Genomics-Assisted Breeding. *Plants*, 14(1), p.142.
2. Das, A.K., **Gowda, M.M.**, Muthusamy, V., Zunjare, R.U., Chauhan, H.S., Baveja, A., Bhatt, V., Chand, G., Bhat, J.S., Guleria, S.K. and Saha, S., 2021. Development of maize hybrids with enhanced vitamin-E, vitamin-A, lysine, and tryptophan through molecular breeding. *Frontiers in plant science*, 12, p.659381.
3. Duo, H., Hossain, F., Muthusamy, V., Zunjare, R.U., Goswami, R., Chand, G., Mishra, S.J., Chhabra, R., **Gowda, M.M.**, Pal, S. and Baveja, A., 2021. Development of sub-tropically adapted diverse provitamin-A rich maize inbreds through marker-assisted pedigree selection, their characterization and utilization in hybrid breeding. *PLoS One*, 16(2), p.e0245497.
4. Pal, S., Zunjare, R.U., Muthusamy, V., Duo, H., **Gowda, M.M.**, Bhowmick, P.K., Kasana, R., Bhatt, V. and Hossain, F., 2020. Influence of T-, C-and S-cytoplasm on male sterility and their utilisation in baby corn hybrid breeding. *Euphytica*, 216(9), p.146.
5. Manjunatha, P.B., Harisha, R., Kohli, M., Naik, P.K., Patel, M.K., Ganesh, B.T., Kumar, S., Ragi, S., Patil, S. and **Gowda, M.M.**, 2023. Exploring the world of mungbean: uncovering its origins, taxonomy, genetic resources and research approaches. *International Journal of Plant & Soil Science*, 35(20), pp.614-635.
6. Pal, S., Zunjare, R.U., Muthusamy, V., **Gowda, M.M.**, Duo, H., Chhabra, R., Kasana, R.K., Maman, S., Katral, A., Bhowmick, P.K. and Hossain, F., 2025. Molecular characterization of T-, C-and S-specific cytoplasmic male sterile (CMS) maize inbreds using mitochondrial SSRs for its utilization in baby corn breeding. *Molecular Biology Reports*, 52(1), p.520.
7. Harisha, R., Bhadr, D., Bhargava, K., Vanisri, S., Shankar, V.G., Balakrishnan, A.P., **Gowda, M.M.** and Singhal, S., 2022. Assessment of variability and genetic diversity for elite rice (*Oryza sativa* L.) genotypes of Telangana and Andhra

Pradesh. *International Journal of Environment and Climate Change*, 12(11), pp.3612-22.

8. Senapati, M., Tiwari, A., Vinod, K.K., Bashyal, B.M., Amaresh, Nagarajan, M., Dhawan, G., **Gowda, M.M.**, Ellur, R.K., Bhowmick, P.K. and Bollinedi, H., 2024. High-density single nucleotide polymorphism-based linkage mapping reveals novel quantitative trait loci for rice (*Oryza sativa*) sheath blight resistance from wild rice (*Oryza rufipogon*)-derived introgression line. *Plant Breeding*, 143(6), pp.828-839.
9. Premkumar, Singh, D., Yadav, J., Chavhan, M., Mohanapure, P., Rath, B., Hada, A., Somvanshi, V.S., KK, V., Bhowmick, P.K., Bollinedi, H. and Dash, M. Gowda, M.M., Ellur, R.K., 2026. RNAseq based comparative gene expression reveals early defense activation and involvement of phenylpropanoid pathway in rice root-knot nematode resistance. *Physiological and Molecular Plant Pathology*, p.103154.
10. **Gowda, M.**, Harisha, R., Bhargava, K. and Adithya, P., 2022. Mutants in starch biosynthesis pathway, characterization and application in corn breeding. *Agric Res*, 3, pp.1-5.

KEY SKILLS

- ❖ Development of molecular markers and their application in plant breeding
- ❖ Familiar with DNA, RNA and Protein extraction and PCR based analysis
- ❖ Knowledge of hybrid breeding and seed production
- ❖ Knowledge of high throughput genotyping platforms (GBS, Microarray)
- ❖ Data Analysis (R, Darwin, ICIM, Haploview, primer3 and TASSEL)
- ❖ Knowledge in Quantitative genetics, Statistics and Bioinformatics
- ❖ GWAS, QTL mapping, MABB and GS